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PALEOMAGNETIC RESULTS FOR EOCENE VOLCANIC ROCKS FROM NORTHEASTERN WASHINGTON AND THE TERTIARY TECTONICS OF THE PACIFIC NORTHWEST

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Abstract. The direction of remanent magnetization for 102 sites in Eocene volcanic and volcaniclastic rocks of the O'Brien Creek Formation, Sanpoil Volcanics, and Klondike Mountain Formation suggests approximately 25° of clockwise rotation of a 100 by 200 km area in northeastern Washington. The volcanic rocks consist chiefly of rhyodacite and quartz latite flows, with intercalated ash flow tuff and volcaniclastic layers. These rocks have been sampled at 102 sites distributed among five volcanotectonic depressions: the Toroda Creek, Republic, Keller, and First Thought grabens and the Spokane-Enterprise lineament. The volcanic rocks probably range in age from 55 m.y. to about 48 m.y., and the 50- to 48-m.y.-old volcanic rocks within this suite appear to be rotated as much as the older rocks. Previous investigators have shown that 40-m.y.-old and younger plutonic rocks of northwestern Washington are not rotated; hence we infer that the north-central Washington rocks were rotated to their present declination between 48 and 40 m.y. B.P. (during the middle and/or late Eocene). During early Eocene time this region was extended in a westward direction through crustal necking, gneiss-doming, diking, and graben formation. Internal deformation of the region related to this crustal extension was extreme, but most bedrock units that were formed concurrent with the crustal extension were probably in place prior to the rotation; hence we infer that the rotation was chiefly accommodated by movement on faults peripheral to the sampled area. Faults active during Paleogene time appear to define boundaries of a triangular crustal block (the Sanpoil block), encompassing much of northeastern Washington, northern Idaho, northwestern Montana, and adjacent parts of British Columbia. The faults include the Laramide thrusts of the Rocky Mountain thrust belt, the strike-slip faults of the Lewis and Clark line, and strike-slip faults of the Straight Creek-Fraser zone. We suggest that during early Eocene time the Sanpoil block was extended westward through crustal necking and dilation and then during the middle Eocene was rotated clockwise and thrust over the craton in a final stage of Laramide thrusting. The "motor" driving these deformations presumably was interaction of North America with oceanic lithosphere off its western margin; such interaction probably involved right-oblique underthrusting and dextral shear.

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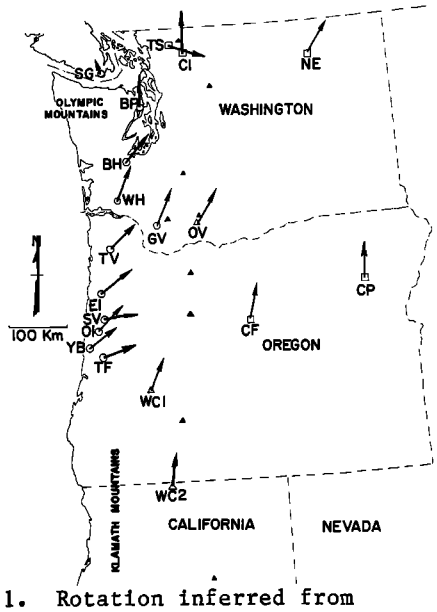


Fig. 1. Rotation inferred from paleomagnetism of Tertiary rocks from the Pacific Northwest. Data referenced in Table 1. Arrows indicate rotation values (not measured declination); study CI shows no rotation, SG shows a rotation of 22.5° counterclockwise, TS a rotation of 110.5° clockwise. For a discussion of rotation statistics, see Beck [1980]. Open circles, Tertiary rocks of Oregon and Washington Coast Ranges; open triangles, Cascade volcanic rocks; squares, other studies; solid triangles, major Cascade volcanoes.

INTRODUCTION

This study was initiated in 1978 as a simple exercise in finding the edge of stable (autochthonous) North America in the Pacific Northwest. Work by Simpson and Cox [1977] and many others (summaries by Magill et al. [1981] and Beck, [1980]) had shown by then that the western edge of the continent, from at least the Klamath Mountains to Vancouver Island and including areas as far east as the Colorado Plateau [Hamilton, 1981], consisted of large crustal blocks that had rotated clockwise relative to North America during early Tertiary time (Figure 1 and Table 1). Together with most investigators we attributed these coastal block rotations to continental margin tectonics associated with accretion and/or dextral shear, and we fully expected to find that the Eocene volcanic rocks of northeastern Washington lay outside the rotated zone. Through an accident of sampling, our first results tended to confirm our prejudices [Schwimmer et al., 1979]. However, the work we report here shows that northeastern Washington has indeed experienced a large clockwise rotation since early Eocene time. This paper contains a summary of the paleomagnetic data as well as some speculations about what they mean. For a complete discussion of sampling and laboratory techniques, a description of individual sampling localities, and a discussion of experimental problems (as well as alternate tectonic interpretations) the reader is referred to the theses of Schwimmer [1981] and Faxon [1982].

GEOLOGIC SETTING

Eocene volcanic rocks within our study area (Figure 2) are patchily distributed over Precambrian and early Paleozoic miogeoclinal rocks on the east and Paleozoic and early Mesozoic eugeosynclinal rocks on the west. The miogeoclinal rocks form westward thickening wedges that include the Belt Supergroup and Windermere Group. The Precambrian sediments within these wedges are overlain by lower Paleozoic shelf deposits.

In the northern part of the study area the eugeosynclinal deposits (Anarchist Group and Mount Roberts Formation) consist of thick stratiform sequences of upper Paleozoic volcanogenic sediments and intercalated lavas, capped by Permian or Triassic (Kobau Formation) and lower Mesozoic marine lavas and marine sediments. In the southern part of the area the eugeosynclinal deposits consist chiefly of a thick sequence of turbidites and intercalated lavas (the Covada Group which is here considered to be Ordovician in age following the usage of Snook et al. [1981]).

The stratified rocks of the miogeoclinal and eugeosynclinal provinces are much folded, faulted, and in places metamorphosed and invaded by granitic rocks. The granitic rocks range in age from Late Triassic to Eocene [Fox et al., 1977] and in much of the eugeosynclinal province are the dominant element of the bedrock. The metamorphic rocks include bodies of lineated, commonly sillimanite-grade gneisses of enigmatic origin referred to as the Okanogan and Kettle

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TABLE 1. Tertiary Paleomagnetic Data for the Pacific Northwest

Study	Unit/Rock Type	Age	Reference
TS	Twin Sisters Dunite	Paleocene(?)	Beck [1975]
CI	Cascade intrusive	Eocene-Miocene	Beck et al. [1982]
NE	northeastern Washington volcanic rocks	Eocene	this paper
SG	East Sooke Gabbro	Oligocene(?)	Symons [1973b]
BP	Bremerton/Point Townsend volcanic rocks	Eocene	Beck and Engebretson [1982]
BH	Black Hills volcanic rocks	Eocene	Globerman et al. [1982]
WH	Willapa Hills volcanic rocks	Eocene	Wells and Coe [1980]
GV	Goble Volcanic Series	Oligocene	Beck and Burr [1979]
OV	Ohanapecosh Volcanics	Oligocene	Bates et al. [1981]
TV	Tillamook Volcanics	Eocene	Magill et al. [1981]
CP	Columbia River Basalt Group	Miocene	Watkins and Baski [1974]
EI	Eocene basaltic sills	Eocene	Beck and Plumley [1980]
SV	Siletz River Volcanics	Eocene	Simpson and Cox [1977]
CF	Clarno Formation	Eocene	C. S. Gromme (manuscript in preparation, 1985)
OI	Oligocene basaltic sills	Oligocene	Beck and Plumley [1980]
YB	Yachats Basalt	Eocene	Simpson and Cox [1977]
TF	Tyee and Flournoy formations (sedimentary)	Eocene	Simpson and Cox [1977]
WC1	Western Cascade volcanic rocks	Oligocene and Miocene	Magill and Cox [1981]
WC2	Western Cascade volcanic rocks	Oligocene and Miocene	Craig [1981]

Locations shown on Figure 1.

gneiss domes and similar rocks of the Tenas Mary Creek sequence of Parker and Calkins [1964]. As used here, the term "gneiss dome" does not imply any particular mode of origin.

Foliation within the gneiss domes is generally conformable with their outer surface; that surface is convex upward with sides sloping away at low but gradually increasing angles. The Okanogan and Kettle gneiss domes are bordered on their west and east sides, respectively, by low-angle shear zones composed of mylonitic rock that locally is several kilometers thick. The gneiss domes, mylonitic border zones, and associated rocks form metamorphic core complexes that are part of a chain of similar complexes scattered along the Cordillera [Fox et al., 1977, p. 18].

The potassium-argon and fission track thermal ages of the lineated gneisses of the Okanogan dome and of the correlative rocks of the Shuswap metamorphic complex to the north (in Canada) indicate that they cooled through the blocking temperatures of biotite, muscovite, apatite, and sphene during the early Eocene [Medford, 1975; Fox et al., 1977]. Gneiss domes and metamorphic core complexes have not as yet been identified within areas west of the study area. To the east, however, the Spokane dome [Cheney, 1980, p. 470], Newport fault zone [Miller, 1971], and the igneous complex of the Selkirk Crest [Miller, 1983] may constitute elements of a large core complex, to which we informally refer as the Spokane-Selkirk core complex. The Newport fault zone is a low-angle mylonite zone some 230 m in width [Miller, 1971] that resembles the mylonite zones bordering the west and east sides of the Okanogan and Kettle domes, respectively. Miller and Engels [1975] traced the Newport fault into northern Idaho, showing that the U-shaped structure (in map view) is synformal with northward plunge. The fault cuts the 48-m.y.-old Silver Point Quartz Monzonite [Miller, 1971], shown as part of the Spokane dome by Cheney [1980]. The fault in places is probably

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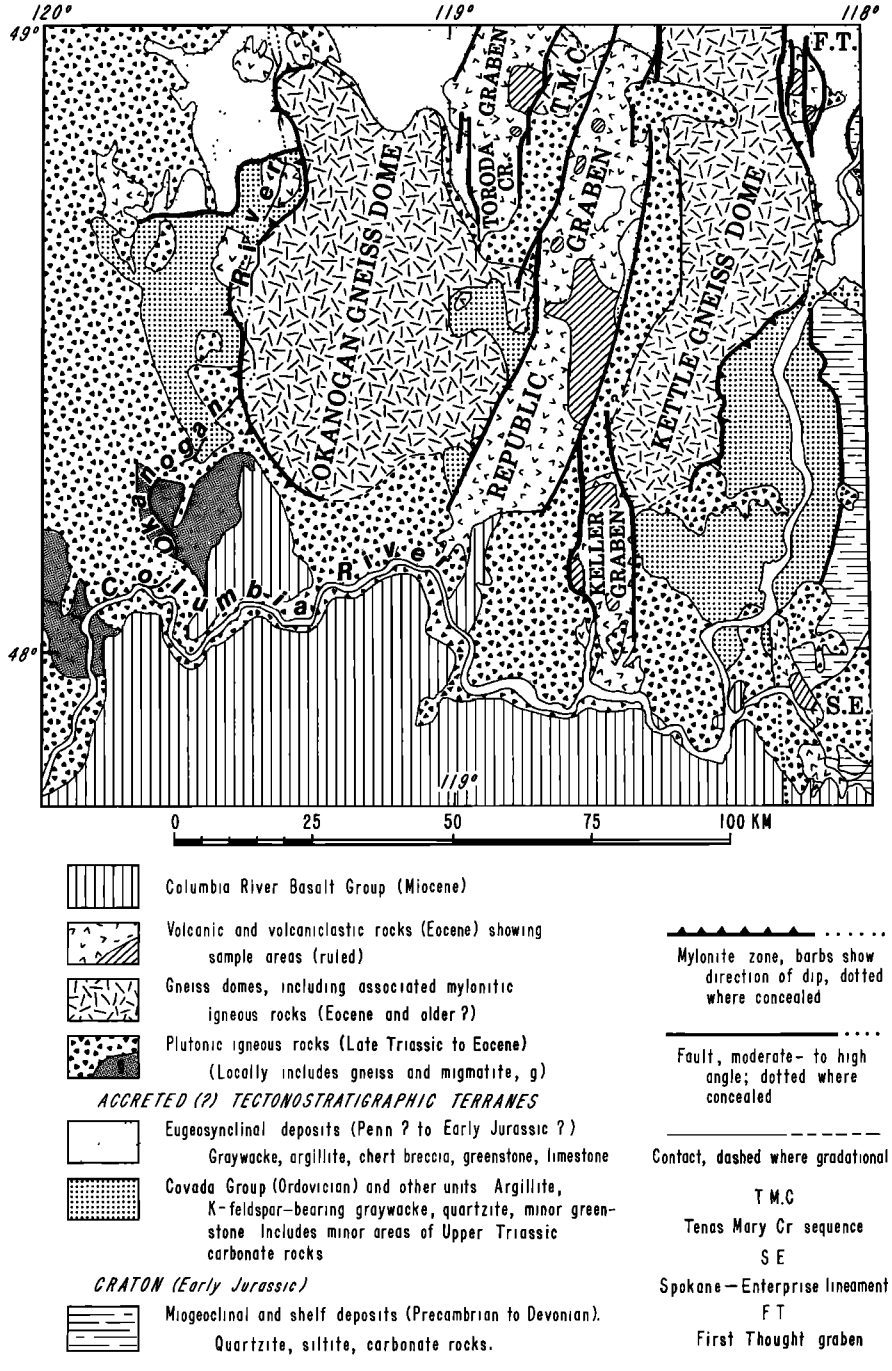


Fig. 2. Generalized geologic map of study area, showing distribution of Eocene volcanic rocks and sampling areas. Geology modified from Fox et al. [1977] and Atwater and Rinehart [1984].

overlain by conglomerate of the Eocene and younger Tiger Formation [Miller, 1971] but elsewhere cuts lower and middle Eocene volcanic and clastic deposits [Harms, 1983]. Tuffaceous rocks at or near the base of the Tiger were correlated with O'Brien Creek Formation, described below, and dated at 54.5±1.5 m.y. [Pearson and

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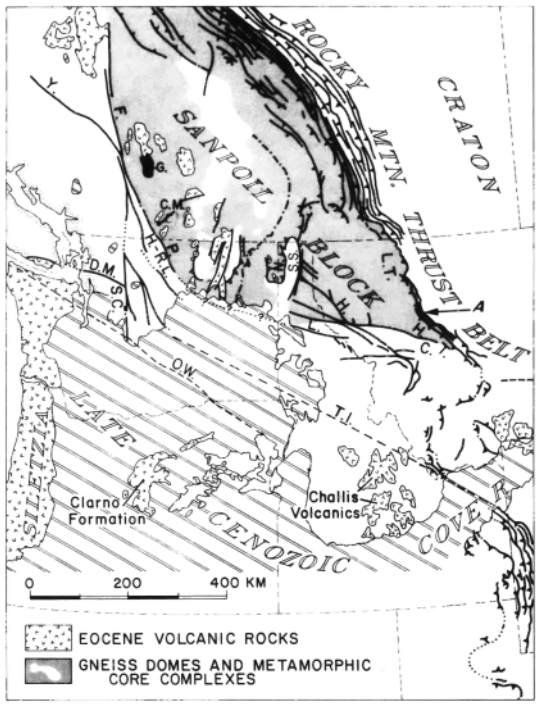


Fig. 3. Eocene volcanic rocks and structures active or possibly active in Paleogene time. Geology modified from King [1969], Miller [1983], Cheney [1980], Fox et al. [1977]. Structures include D. M. (Devils Mountain fault), S. C. (Straight Creek fault), T. I. (Trans-Idaho discontinuity), O. W. (Olympic-Wallawa lineament). Other features mentioned in text include L. T. (Lewis thrust), H. T. (Hoadley thrust), P. (Pasayten fault), H.-R. L. (Hozameen-Ross Lake fault), Y. (Yalakom fault), F. (Fraser fault), L. C. (Lewis and Clark line), N. (Newport fault), H. (Hope fault), S. S. (Spokane-Selkirk core complex), G. (Guichon batholith), and C. M. (Copper Mountain intrusion). Point labeled A is pivot point of Lewis and Hoadley thrust faults [Mudge and Earhart, 1980]. Dot-dash line is contact between Quesnellia (west) and craton (east).

Obradovich, 1977]. Hence the Newport fault was active during the early and middle Eocene.

The discordance of the paleomagnetic direction of the Eocene volcanic rocks, the central finding of this study, could be caused by rotation in a horizontal plane of the crustal block or blocks in which the volcanic rocks are embedded. Nearly the entire western margin of the North American continent is believed to be a collage of such crustal blocks, including blocks that are allochthonous and that were accreted to the craton during Mesozoic and early Tertiary time [Beck et al., 1980; Monger et al., 1982].

Early Tertiary structural features (Figure 3) which could be related to the tectonic rotation inferred from the discordance of the Eocene paleomagnetic poles include the thrust faults of the Rocky Mountain fold and thrust belt, the strike-slip faults of the Lewis and Clark line [Reynolds, 1977; Reynolds and Kleinkopf, 1977], the left-lateral Devils Mountain fault zone [Whetten, 1978], the right-lateral Straight Creek fault zone [Misch, 1977], and also certain more enigmatic features, including the Olympic-Wallowa lineament [Raisz, 1945] and Trans-Idaho discontinuity [Yates, 1968].

EOCENE VOLCANIC ROCKS

The volcanic rocks we sampled are remnants of a volcanic field that probably extended throughout much of south-central British Columbia and northeastern Washington. Pearson and Obradovich [1977] showed that the stratified rocks of this volcanic field are divisible into three stratigraphic units (originally recognized within the Republic graben by Muessig [1962]). They are, in ascending order, the O'Brien Creek Formation (terrigenous sediments and crystal-lithic tuff), the Sanpoil Volcanics (porphyritic lithoidal lava flows, and subordinate porphyritic glassy lava flows, volcanic breccia, and tuff), and the Klondike Mountain Formation (laharic deposits and tuffaceous sediments at the base and aphyric lithoidal lava flows and glassy lava flows above). Except for the basal subunits of the Sanpoil, the lavas of the Sanpoil and Klondike Mountain Formation range in silica content from about 59 to 69% and chemically are dark rhyodacite, rhyodacite, dark quartz latite, and dacite (classification of Rittman [1952]). The basal part of the Sanpoil in the Toroda Creek graben, and the correlative of the Sanpoil in British Columbia, the Marron Formation, consists of alkalic rocks that include trachyte [Pearson, 1967], analcite-lava and rhomb porphyry [Daly, 1912, p. 98], and sodic trachytes and phonolites [Monger, 1968]. As might be expected from the presence of the alkalic end members, the suite as a whole is alkalic-calcic (classification of Peacock [1931]).

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The O'Brien Creek Formation is as much as 1.3 km in thickness in the Republic graben [Muessig, 1967, p. 46]. Correlative strata in Canada are 120 m in thickness [Monger, 1968]. The Sanpoil Volcanics probably reaches a thickness of 2.4 km in the Toroda Creek graben [Pearson, 1967], and we measured a partial section of 2.2 km at its type section in the Republic graben. The Sanpoil correlative in Canada, the Marron Formation, is 1.8 km in thickness [Monger, 1968]. The Klondike Mountain Formation reaches its maximum thickness, estimated to be 1.2 km, in the Toroda Creek graben [Pearson and Obradovich, 1977, p. 27].

Hypabyssal dikes cut all the volcanic rocks except the youngest aphyric glassy lava flows of the Klondike Mountain Formation. Most of the dikes dip steeply and strike parallel to the grabens. They also cut bedrock outside the grabens, including the gneiss domes, and in places are so abundant as to constitute sheeted dike swarms. Moye et al. [1982] estimated that in places near the boundary faults of the grabens up to 4 km of extension is required to accommodate the dikes. Adjacent to and east of the Republic graben, the dikes form over 90% of the bedrock in zones several kilometers wide (R. W. Holder, personal communication, 1982). Epizonal and mesozonal plutons similar in age to the volcanic rocks are widespread, both in the study area and in nearby parts of British Columbia.

The O'Brien Creek Formation, Sanpoil Volcanics, and Klondike Mountain Formation are early and middle Eocene on the basis of K-Ar age determinations as follows: One sample of biotite from bedded tuff correlated with the O'Brien Creek yielded a K-Ar age of 54.5 ± 1.5 m.y. [Pearson and Obradovich, 1977, p. 13]. Eleven mineral separates, including plagioclase, biotite, and hornblende, from seven samples of extrusive phases of the Sanpoil ranged in age from 49.6 ± 3.0 to 53.8 ± 1.8 m.y., and three determinations on hornblende and three on biotite from two samples of the Klondike Mountain ranged in age from 47.5 ± 1.7 to 50.4 ± 1.7 m.y.; a third sample of the Klondike Mountains yielded hornblende ages of 42.4 ± 2.0 and 42.5 ± 1.5 m.y. [Pearson and Obradovich, 1977] (ages revised to International Union of Geological Sciences (IUGS) constants using correction factors as given by Dalrymple [1979]). The age of the third sample is suspect because it was collected from a flow below the other two. A sample from rocks provisionally correlated with the Sanpoil Volcanics in the Keller graben yielded a hornblende age of 50.3 ± 0.38 m.y. [Atwater and Rinehart, 1984].

Four K-Ar ages, representing two plugs and two flows in the Okanogan Valley, range from 53.4 ± 2.3 to 46.3 ± 2.0 m.y. [Rinehart and Fox, 1972; p. 63; 1976, p. 40] (ages revised to new IUGS constants using correction factor as given by Dalrymple [1979]). Yates and Engles [1968] reported the K-Ar age of a hypabyssal intrusive rock in the Williams Lake area correlated by them with the Sanpoil to be 51.1 ± 1.5 m.y.

In Washington the remnants of this volcanic field are largely but not entirely confined to a family of echelon north-northeast trending volcano-tectonic depressions (Figure 2). The larger of these are the Toroda Creek graben, Republic graben, Keller graben, and a linear feature referred to as the Spokane-Enterprise lineament [Becraft and Weiss, 1963]. In British Columbia, coextensive or correlative volcanic rocks have wider distribution, forming the bedrock throughout extensive areas. According to Souther [1970], the eruption of these volcanic rocks was accompanied by block foundering, north-south block faulting, formation of large cauldron subsidence features, and emplacement of north-south dike swarms. These rocks are also Eocene [Mathews and Rouse, 1963; Mathews, 1964].

The Eocene volcanic rocks reviewed above are part of a broad south-southeast trending belt of Eocene extrusive rocks that extends into eastern Idaho and southwestern Montana. The belt includes most of the Challis Volcanics of central Idaho and the Absaroka Volcanic Supergroup of former usage of south-central Montana and northwestern Wyoming. The Clarno Formation of north-central Oregon (Figure 3) may be also part of this belt, though on the basis of its known age range (41-46 m.y. [Noblett, 1981]), it is younger than the otherwise similar volcanic rocks referred to above. The belt has been referred to as the Challis arc [Vance, 1979].

PALEOMAGNETISM

Field and Laboratory Procedure, and Rejection Criteria

In four field seasons we sampled 126 lava flows, ash flow tuffs, sedimentary units, dikes, and sills from six general areas in northeastern Washington. Most of

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the units sampled are exposed within the five distinct structural depressions shown on Figure 2. The exceptions are a few intrusive units sampled near the boundary faults of the major grabens and used for polarity studies only; these were not included in directional calculations. Sampling involved standard paleomagnetic techniques, using portable core drilling equipment. In general, we took seven separately oriented samples per site, spacing the samples several meters or more apart. "Sites" were chosen to represent "points in time," as nearly as possible; that is, we tried to insure that all samples from a single site acquired their directions of remanent magnetization with a few years of one another. Individual flows or ash flow units and thin sedimentary layers were generally chosen as sites. We took care to avoid placing two or more sites in the same flow or sedimentary unit (thereby giving undue weight to that particular "sample" of the paleogeomagnetic field). However, the structural complexity of the area probably assures that we have double-sampled in a few cases.

All samples were alternating field demagnetized, using a two-axis reciprocating tumbler. The method used to determine optimum demagnetization level involved examination of orthogonal plots for two or more pilot samples for each site as well as use of the minimum dispersion criterion [Irving, 1964]. Examples are given by Schwimmer [1981] and Faxon [1982].

As in most paleomagnetic studies, we sometimes encountered sites that grouped poorly (or not at all) despite magnetic cleaning. In order to exclude poorly determined or spurious directions from the data set we arbitrarily discarded any site for which the circle of 95% confidence [Irving, 1964] exceeded 15°. Only five sites failed this test. More common, however, was the situation in which a single sample gave a direction that fell well outside the cluster defined by the remainder of the site. Such cases probably represent gross orientation errors or inadvertent sampling of blocks that are not strictly in situ. To deal with this situation objectively, we discarded the sample if its direction diverged from the mean by an angle exceeding twice the within-site angular standard deviation [Irving, 1964]; the mean direction and 95% confidence interval then were recalculated before the 15° cutoff was applied. Thirty-two sites had their directions slightly modified by this procedure. Schwimmer [1981] and Faxon [1982] give details of within-site dispersions and rejection procedures.

Data, Statistical Analysis, and Problems

Excluding sites for which the circle of 95% confidence exceeds 15° and sites (intrusions) for which the structural correction is unknown, we have 102 sites upon which to base our mean direction. These are predominantly lavas of the Sanpoil Volcanics; the distribution of sites and their polarities are Klondike Mountain Formation: 12 sites, reverse polarity; Sanpoil Volcanics: 85 sites, normal polarity; O'Brien Creek Formation: three sites, mixed polarity; and O'Brien Creek Formation-Sanpoil(?) Volcanics: two sites, normal polarity. Our sampling thus spans at least two polarity transitions and almost certainly includes enough time to average out the geomagnetic secular variation.

Tables 2 and 3 summarize our results, which are plotted in Figure 4. Owing to the availability of exposures sampling was very unevenly distributed; thus in Table 3 the data are divided into six very unequal subsets (sampled areas are shown on Figure 2). The mean direction, calculated in two different ways, is shown near the bottom of Table 3, for comparison with the appropriate expected direction for northeastern Washington shown in the last row, calculated from Irving and Irving [1982].

The method of calculating the mean requires some comment. The usual procedure is to assign unit weight to each site-mean direction on the grounds that each direction is an independent spot reading of a direction of the ancient geomagnetic field. This may not always be the case, however, and it is particularly likely to be in error for thick piles of volcanic rock erupted over an unknown (and possibly short) period of time. Thus it could be argued that the 33 lavas sampled in the southern part of the Republic graben (Table 3) were erupted in such rapid sequence that they represent only 15, or 5, or even one, truly independent spot reading of the Eocene geomagnetic field. In the absence of evidence proving accumulation over a long period of time (soil horizons, polarity transitions, or unconformities, for instance), such doubts must always exist. Row 8, Table 3, takes the extreme view that each sampled area represents only a single determination of

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TABLE 2. Directions of Remanent Magnetization for Eocene Volcanic Rocks of Northeastern Washington After Magnetic Cleaning and Tectonic Correction

Site	N	D,deg	I,deg	k	$\alpha 95$	DAZ	DPL	$\tilde{H}$	λ,deg	ϕ,deg
K1	3	249.6	-52.7	39.9	12.8	--	0	30	48.806	118.683
K2	7	244.8	-74.4	138.3	4.5	--	0	30	48.806	118.683
K3	6	184.1	-65.0	266.7	3.5	90	35	45	48.877	118.786
K4	4	186.4	-42.5	29.9	12.8	97	5	30	48.877	118.786
K5	8	173.4	-48.4	48.6	7.1	97	5	30	48.877	118.786
K6	7	128.3	-66.7	48.5	7.6	97	5	60	48.877	118.786
K7	7	129.5	-67.6	529.3	2.3	97	5	60	48.875	118.808
K8	8	182.1	-51.4	40.3	7.8	97	5	50	48.875	118.808
K9	8	202.3	-53.0	56.2	6.6	88	22	100	48.829	118.835
K10	7	217.1	-59.6	77.8	6.0	106	15	45	48.829	118.835
K11	9	241.4	-57.5	98.6	4.7	--	0	20	48.647	118.739
K12	6	243.9	-27.3	194.3	4.1	97	5	100	48.877	118.786
KM	(12)	202.0	-62.1	11.4	13.4	--	--	--	--	--
S1	7	317.2	53.1	175.0	4.0	--	0	20	48.237	118.673
S2	6	320.1	64.1	339.9	3.1	--	0	40	48.237	118.673
S3	6	315.9	66.3	617.5	2.3	--	0	40	48.237	118.673
S4	7	317.1	62.1	700.0	2.0	--	0	40	48.237	118.673
S5	4	318.1	66.2	400.0	3.5	--	0	40	48.237	118.673
S6	6	330.2	66.2	483.2	2.6	--	0	30	48.237	118.673
S7	6	59.7	84.3	31.4	10.2	--	0	40	48.237	118.673
S8	7	272.3	80.3	273.4	3.2	--	0	40	48.237	118.673
S9	6	358.5	67.1	82.3	6.3	--	0	30	48.237	118.673
S10	6	15.3	55.9	168.7	4.4	--	0	40	48.237	118.673
S11	8	322.1	68.5	200.0	3.5	248	45	35	48.524	118.616
S12	9	329.0	67.6	150.8	3.8	248	45	30	48.524	118.616
S13	8	328.2	48.1	169.7	3.8	251	47	50	48.524	118.616
S14	8	329.3	80.0	90.6	5.2	258	50	30	48.527	118.629
S15	4	50.0	81.4	102.9	6.9	260	50	30	48.527	118.629
S16	5	235.8	87.7	185.3	4.6	260	50	35	48.530	118.642
S17	6	301.9	71.1	339.9	3.1	266	53	30	48.530	118.642
S18	7	327.6	66.2	700.0	2.0	271	54	35	48.530	118.642
S19	5	6.8	64.9	286.3	3.7	275	54	35	48.533	118.666
S20	7	351.3	44.6	132.3	4.6	277	53	35	48.533	118.666
S21	7	81.7	71.7	273.4	3.2	218	35	25	48.573	118.640
S22	9	36.7	60.4	56.7	6.2	298	40	70	48.004	118.653
S23	8	30.2	53.1	76.8	5.6	330	30	50	48.004	118.553
S24	8	29.9	83.3	13.7	12.0	250	15	80	48.143	118.740
S25	7	50.8	71.9	86.0	5.7	250	15	95	48.143	118.740
S26	6	352.3	70.8	67.6	7.0	250	15	50	48.143	118.740
S27	7	312.8	63.2	31.2	11.0	250	15	70	48.143	118.740
S28	7	334.1	64.2	39.5	9.7	250	15	70	48.143	118.740
S29	7	31.1	47.0	172.3	4.6	272	15	70	48.375	118.731
S30	5	220.8	74.1	50.2	8.8	230	45	90	48.375	118.731
S31	8	249.7	71.9	257.4	3.5	230	45	30	48.375	118.731
S32	6	230.6	68.7	117.6	5.3	230	45	50	48.375	118.731
S33	7	240.0	71.5	354.9	2.8	230	45	40	48.375	118.731
S34	7	341.0	89.0	483.3	2.4	280	50	60	48.533	118.750

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TABLE 2. (continued)

Site	N	D,deg	I,deg	k	α 95	DAZ	DPL	$\tilde{H}$	λ,deg	ϕ,deg
S35	6	301.7	63.3	33.0	11.8	174	70	70	48.402	118.729
S36	7	352.5	64.2	56.8	7.0	250	23	70	48.468	118.639
S37	8	213.7	78.2	246.1	3.2	191	31	50	48.755	118.125
S38	10	52.1	48.3	41.8	7.6	30	25	50	48.749	118.126
S39	6	3.6	45.9	228.9	4.4	302	40	20	48.470	118.725
S40	6	281.4	71.1	255.0	4.2	302	40	70	48.507	118.736
S41	7	82.8	58.9	180.1	4.5	70	30	30	48.324	118.704
S42	6	37.2	46.7	781.2	2.4	70	30	30	48.323	118.709
S43	7	32.4	52.6	214.3	4.1	70	30	30	48.322	118.709
S44	7	34.7	50.6	857.1	2.0	70	30	15	48.321	118.717
S45	7	44.7	53.7	857.1	2.0	70	30	30	48.319	118.720
S46	7	49.3	46.1	315.8	3.4	70	30	15	48.317	118.721
S47	7	51.4	52.4	127.7	5.4	70	30	20	48.310	118.724
S48	7	34.3	45.1	375.0	3.2	70	30	30	48.310	118.725
S49	7	28.3	42.5	741.5	2.5	70	30	30	48.309	118.726
S50	7	57.0	39.1	987.9	1.9	50	10	20	48.339	118.722
S51	7	72.3	61.0	309.3	3.4	50	10	30	48.340	118.725
S52	7	42.5	51.9	670.9	2.3	50	10	15	48.339	118.726
S53	7	36.0	47.2	1,093.8	1.8	50	10	20	48.338	118.726
S54	7	37.6	52.2	1,172.8	1.8	50	10	15	48.336	118.726
S55	7	2.1	53.0	229.0	4.0	--	0	20	48.488	118.808
S56	7	50.4	57.3	488.6	2.7	--	0	60	48.484	118.803
S57	7	347.9	80.5	234.9	4.0	--	0	100	48.486	118.789
S58	7	44.1	68.7	290.9	3.5	60	10	20	48.341	118.708
S59	7	12.9	72.6	552.5	2.6	60	10	30	48.343	118.717
S60	7	317.7	59.0	622.3	2.4	60	10	80	48.352	118.736
S61	6	316.2	72.2	531.7	2.9	208	17	80	48.288	118.741
S62	6	302.1	87.1	91.0	8.0	208	17	30	48.288	118.739
S63	6	8.7	86.9	1,087.0	2.0	--	0	60	48.423	118.721
S64	6	333.7	74.6	904.2	2.2	--	0	60	48.337	118.734
S65	7	78.7	45.6	20.9	13.9	55	10	60	48.334	118.702
S66	7	42.7	47.3	128.4	5.4	50	10	40	48.358	118.707
S67	7	105.2	61.5	20.0	14.0	--	0	40	48.359	118.705
S68	7	17.2	69.0	151.8	4.9	--	0	30	48.342	118.707
S69	6	263.2	87.1	94.9	6.9	125	50	30	48.243	118.836
S70	6	65.6	55.9	444.8	2.7	135	45	65	48.234	118.661
S71	7	28.6	38.3	17.5	14.8	225	15	30	47.918	118.167
S72	9	20.7	44.5	50.6	7.3	310	10	30	47.918	118.168
S73	7	43.6	58.3	282.6	3.6	--	0	40	47.918	118.168
S74	7	11.7	57.1	20.0	13.8	257	27	30	47.920	118.170
S75	7	30.5	51.7	112.4	5.7	257	27	30	47.920	118.170
S76	7	20.0	44.0	104.8	5.9	60	48	20	47.950	118.208
S77	9	5.9	53.4	289.1	2.7	60	48	60	47.950	118.206
S78	8	34.6	41.4	55.2	7.5	60	61	20	47.950	118.204
S79	5	0.6	42.7	176.0	5.8	60	61	40	47.950	118.204
S80	8	350.2	50.0	168.0	4.3	327	42	30	47.910	118.167
S81	8	32.3	82.8	82.3	6.1	255	26	30	47.908	118.165

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TABLE 2. (continued)

Site	N	D,deg	I,deg	k	α_{95}	DAZ	DPL	$\tilde{H}$	λ,deg	ϕ,deg
S82	7	182.4	79.5	312.4	3.4	225	21	30	47.905	118.162
S83	8	293.5	77.2	114.4	5.2	252	35	40	47.901	118.160
S84	5	327.9	54.4	47.1	11.3	282	47	30	47.900	118.160
S85	9	350.5	77.2	144.5	4.3	270	21	20	47.892	118.158
SM	(85)	13.6	69.5	12.4	4.6	--	--	--	--	--
01	9	53.0	71.7	167.8	11.4	225	31	60	48.524	118.600
02	8	192.6	-54.5	101.6	4.9	35	30	50	48.125	118.740
03	7	7.5	57.1	31.5	9.4	30	25	30	48.829	118.131
OS1	9	348.8	63.1	52.2	7.2	290	10	40	48.859	118.356
OS2	7	46.4	48.1	120.8	4.8	332	55	50	48.524	118.602
OOSM	(5)	34.3	60.8	28.0	14.7	--	--	--	--	--

The initial letters under site give lithology: K, Klondike Mountain; S, Sanpoil; O, O'Brien Creek; OS, two flows that are either Sanpoil or O'Brien Creek. KM, SM, and OOS designate means for the Klondike Mountain Fm., Sanpoil Volc., and O'Brien Creek Fm., plus O'Brien Creek Fm.-Sanpoil(?) Volc. sites, respectively. N is the number of samples (number of sites, for mean calculations). D, I are declination and inclination, after magnetic cleaning and tectonic correction. k is the estimate of Fisher's [1953] precision parameter. α_{95} is the radius of confidence at the 95% probability level. DAZ, DPL are the azimuth and plunge of the dip vector, respectively. $\tilde{H}$ is the peak value of the alternating magnetic field used to clean the site, in milliteslas. λ, ϕ are the north latitude and west longitude of the sampling site, respectively.

the direction of the Eocene field. The results of this "localities" calculation, however, are virtually identical to the alternative (row 7, Table 3) in which the site-mean directions are used as data. Thus in this case the question is relatively unimportant. The "site-mean" calculation is almost certainly preferable, however, for the following reasons: (1) The within-area dispersions for all six areas are of the same order of magnitude as the dispersion of the geomagnetic field (i.e., Creer, 1962; Cox, 1962, 1970). The between-area dispersion, on the other hand, is much lower, and in fact is of the order of what one would expect from a set of independent estimates of a single paleomagnetic direction. If the flows at each sampled area represented very short periods of time the picture should be reversed; within-area dispersions would be small and the between-area dispersion should be of the order of typical geomagnetic dispersions (perhaps 15-25°). (2) None of the six area-mean directions are statistically distinct at the 95% confidence level, as one would expect if they represented well-determined spot estimates of a geomagnetic field showing the usual amount of variability.

The main problem with our data is the perennial and intractable one besetting all paleomagnetic studies of deformed volcanic rocks, finding the proper tilt correction. As is well known, volcanic rocks are rarely distinctly layered, and what layering may happen to exist (flow parting, reddened flow boundaries, trend of vesicular zones, coarse "layering" defined by the tops and bottoms of columnar zones, etc.) may not accurately represent the true paleohorizontal. Interbedded tuff and sediment horizons and compression-layering in ash flow tuffs are much more reliable but usually hard to find. In addition, there is the problem of original dip (as on the side of a volcano). As with most such studies, we attempted to deal with the tilt correction problem by a straightforward, brute force technique: (1) We tried to concentrate our sampling in relatively simple sections where sedimentary or ash flow tuff horizons were available to indicate attitude. (2) We interpolated between

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TABLE 3. Paleomagnetic Data for Eocene Rocks From Northeastern Washington

Group	N	D,deg	I,deg	Δ,deg	α_{95},deg	Polarity
Toroda Creek	10	187.4	-63.8	27.7	18.4	R
N Republic	21	4.4	73.2	19.6	8.1	mixed (19N, 2R)
S Republic	33	34.7	68.5	24.6	8.0	N
Keller	20	353.1	70.3	17.6	7.6	mixed (19N, 1R)
First Thought	3	31.8	70.7	24.8	49.7	N
Spokane	15	13.0	60.1	21.1	10.6	N
Mean, by site	102	15.6	68.9	20.6	3.7	
Mean, by locality	6	14.8	68.3	6.8	6.2	
Expected direction	349.4	69.4		4*		

N, number of sites; D, I, mean declination and inclination; Δ, angular standard deviation; α_{95}, radius of circle of 95% confidence. Expected direction calculated from the 50 m.y. B.P. pole of Irving and Irving [1982].
*Circle of 95% confidence about the mean pole.

attitudes as little as possible. (3) In instances where we were forced to use indicators in the flows themselves as evidence of attitude, we averaged several such readings over nearby outcrops. Only if the "attitude" was fairly consistent from outcrop to outcrop did we accept the determination. (4) We scattered our sampling over as wide an area as possible, thereby maximizing the chance that random errors (and original dip) have averaged to zero. Despite these precautions some of our tilt corrections do not inspire great confidence. Nevertheless, we feel certain that the mean directions given in Table 3 are reliable. Over 100 sites scattered over so wide an area will certainly have averaged any original dip nearly to zero, and the vast majority of sites we used have attitudes that can be characterized as ranging from "good" to "certain." Moreover, scatter decreases when the tilt corrections are applied (angular standard deviation drops from 25.6 to 22.0). This change is in the right direction, although not quite significant statistically, at the 95% confidence level. The mean direction (by site) for the entire data-set without tilt corrections is D, 26.6°; I, 67.0°; α_{95}, 4.7°.

If northeast Washington had remained rigidly attached to the North American craton since the Eocene, the magnetic direction one would expect to find in our rocks is D_x, 349.1°; I_x, 69.4°. This was calculated from the 50 m.y. B.P. reference pole of Irving and Irving [1982] on the assumption that the Eocene magnetic field was that of a geocentric dipole. Thus the mean declination observed in these rocks points clockwise of the expected direction by 26.2°; the 95% confidence limit is 12.2°. The expected and observed inclinations are essentially identical. We conclude that northeastern Washington itself has rotated clockwise some 25° in the past 50 m.y., relative to the North American interior. Because the expected and observed inclinations are nearly identical, this rotation apparently took

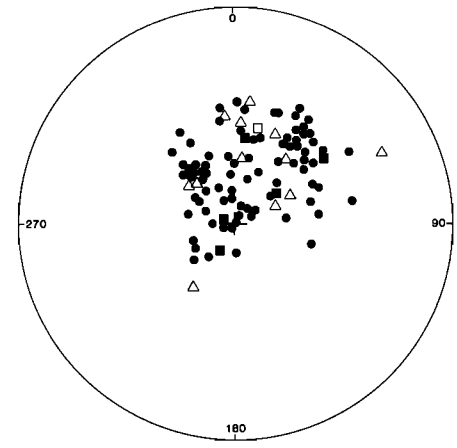


Fig. 4. Mean directions of magnetization for 102 sites from Eocene volcanic rocks of northeastern Washington. Solid symbols indicate normal polarity; open symbols are reverse polarity directions inverted through the center of the projection. Circles, Sanpoil Volcanics; triangles, Klondike Mountain Formation; squares, O'Brien Creek Formation.

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place without any large change in relative latitude.

INTERPRETATION

Timing of rotation. Since the magnetic declination in the Klondike Mountain Formation (line KM, Table 2, the equivalent normal polarity direction is D, 22°; I, 62.1°) is at least as easterly as the mean declination for the data set as a whole, the rotation appears to have occurred sometime after the Klondike Mountain flows were extruded 48-50 m.y. ago. Adjacent Miocene basalts in the Columbia Plateau province are essentially unrotated (Beck [1980] summarizes extensive work by N. D. Watkins and co-workers) except within the crest-hinge area of anticlines [Reidel et al., 1984]. Likewise, the North Cascade block to the west contains plutons as old as 40 m.y. B.P. that are unrotated and in situ [Beck et al., 1982]. Thus if the rotation was truly "regional," it must have occurred between 50 and 40 m.y. B.P. Alternatively, rotation may simply have been confined to the local northeastern Washington area. In the latter case rotation could be much younger than 40 m.y. ago, but a tectonic boundary of some sort must exist west (and south?) of the rotated block. This boundary could be sharp or gradational.

Internal deformation of rotated domain. Extensional features of the Eocene age within the study area include (1) the fault-bounded grabens and associated dikes, and (2) the family of gneiss domes and the similarly lineated and mylonitized rocks of other core metamorphic complexes to the east and north.

The Toroda Creek, Republic, Keller, and First Thought grabens and other volcanotectonic depressions along with their associated dike swarms indicate significant extension of the Sanpoil block in a direction orthogonal to them, now about N70°W.

The terminal mobilization and unroofing of the core metamorphic complexes, judging from their K-Ar and fission track thermal ages, were approximately concurrent with extrusion of the volcanic rocks and formation of the grabens. The complexes, as they appear in map view, are elongated about N20°E, the same as the grabens, probably implying that necking and dilation of the rigid upper crust over the complexes also contributed to the N70°W extension of the Sanpoil block. Price [1980, p. 77] referred to the development of the gneiss domes as a process of crustal scale megaboudinage, associated with lithospheric stretching and clockwise rotation. We agree but stress that the masses of relatively undeformed upper crustal rock between or alongside the gneiss domes are the megaboudins and that the gneiss domes are the ductile matrix [cf. Davis and Coney, 1979].

Also present are batholithic igneous rocks of Eocene age. These include hornblende-biotite granitic rocks that probably are intrusive equivalents of the Eocene volcanic rocks, and biotite-muscovite or biotite granite associated with and locally grading into the lineated rocks of the gneiss domes. The area occupied by these rocks is locally substantial.

The mylonite zones bordering the gneiss domes represent the zone of rheologic transition between megaboudins (country rock) and ductile lithosphere (gneiss domes) in our view or a later distributed thrust of regional extent [Snook, 1965; Cheney, 1979]. The internal fabric of mylonite and crosscutting ultramylonite within the domes record at least two pulses or episodes of slightly nonhomoaxial movement [Wilson, 1981, p. 60]. The most recent mylonitization within and at the exterior of the domes postdates the intrusion of 48-m.y.-old plutons into the developing gneiss domes [Atwater, 1985] and hence the extrusion of the coeval uppermost Klondike Mountain lavas. Late movement on listric faults that merge with and cut the bordering mylonite zones certainly postdates these lavas [Rhodes and Cheney, 1981], but the relative age of that movement and the late mylonitization within the domes is unknown.

Could the rotations reported here chiefly reflect deformation within and between the grabens (deformation, that is, essentially internal to the study area)? The age relations noted above indicate the following sequence of events during Eocene time: (1) Basin development signaling the beginning of crustal necking, and deposition of O'Brien Creek sediments, (2) extrusion of Sanpoil lavas and initiation of graben formation, (3) deposition of lower part of the Klondike Mountain concurrent with uplift of the basins situated over the (growing) gneiss domes,

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(4) intrusion of all older rocks by dikes in part coeval with extrusion of youngest Klondike Mountain lavas, and (5) final downfaulting of grabens, coeval with uplift and possibly coeval with late mylonitization within the interior of the gneiss domes. Our data indicate that the youngest Klondike Mountain rocks are rotated; hence we conclude that rotation followed most but probably not all of the regional dilation. By late Klondike Mountain time all major bedrock units were in place, hence we do not think we have room to horizontally rotate the grabens as individual elements some 25° or so.

The grabens are tied to adjacent crystalline terranes, including especially the Kettle gneiss dome (see map of Pearson [1977]) by the concentration of dike swarms within the grabens and areas adjacent to the graben. Hence it does not seem likely that the grabens are remnants of a downfolded (and rotated) upper plate that was measurably displaced along a basal decollement from the source area of the volcanic rocks [cf. Cheney, 1979].

We tentatively conclude that our study area was rotated en block and that the rotation was accommodated by movement on faults outside this area.

External boundaries of rotated domain. The early Tertiary clockwise rotation suggested by our data implies that the boundary between the collage of tectonically rotated blocks composing the western margin of the North America plate and the craton lies between northeastern Washington and the unrotated Eocene rocks of central Montana (localities 17, 30, 31, and 32 of Irving and Irving [1982]. We also suppose that there are tectonic boundaries between northeastern Washington and other elements of this collage to the west and south. However, we know of no major Eocene or younger strike-slip faults within the 100 by 200 km area spanned by our sample sites or, for that matter, between our area and the Late Triassic Guichon batholith and Copper Mountain Intrusion of southern British Columbia, both of which show substantial clockwise rotation [Symons, 1971, 1973].

East of the study area, the potassium-argon ages of muscovite and biotite from localities within the Spokane-Selkirk core complex indicate that most parts of this complex had cooled through the blocking temperatures of these minerals about 46-54 m.y. B.P. [Miller and Engels, 1975] (ages revised to new constants). Thus the uparching and cooling of the Spokane-Selkirk core complex and extension along the Newport fault were broadly concurrent with uparching and cooling of Okanogan and Kettle gneiss domes, the formation of grabens, and with the extrusion of Sanpoil and Klondike Mountain volcanic rocks. Lacking a specific measure of Eocene crustal extension across the Spokane-Selkirk axis, it is not possible to say whether or not that extension could accommodate the rotation of the Eocene volcanic rocks of our study area. However, we believe that most of the Spokane-Selkirk core complex had formed by late Klondike Mountain time (about 48 m.y. B.P.) hence that the crustal extension represented by deformation and flow of rocks within this complex largely predated the rotation of the Eocene volcanic rocks in our area to the west.

Structures peripheral to our study area that were active in Paleogene time are shown in Figure 3. The rotated domain appears to be bounded on the north and east by either the Rocky Mountain trench or the Rocky Mountain thrust belt and on the west by the Pasayten and Straight Creek faults. The southern boundary, if there is one, is concealed by the Columbia River Basalt Group of Miocene age. Conceivably, faults of the Lewis and Clark line could extend westward below the basalt and form this southern boundary.

The rotated domain, which for convenience may be referred to as the Sanpoil block, includes parts of Quesnellia [Monger et al., 1982] and the pre-Cretaceous craton. The accretion of Quesnellia to the craton was completed prior to the Late Cretaceous; hence that event was probably not a factor in the subsequent (Eocene) rotation of the Sanpoil block.

If the structures enumerated above do form the perimeter of the Sanpoil block, their movement history should provide information about the rotation itself. The faults defining the Idaho and westernmost Montana part of the Lewis and Clark line are echelon, west-northwest trending tear faults with apparent right-lateral displacements of 13-29 km [Harrison et al., 1974, p. 12]. According to Harms [1983], right-lateral displacement on this zone and on the associated Hope fault zone was related to crustal stretching recorded by the Newport fault. The Hope fault splays away to the northwest from the Lewis and Clark line.

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Tension fractures fanning northward from the Hope fault are filled by dacite porphyry and granodiorite dikes dated at about 50 m.y. B.P. [Harrison et al., 1972]. Right-lateral movement of 5-7 km on the Hope fault was in part coeval with intrusion of the dikes [Harrison et al., 1972]. The Hope fault is apparently interrupted by, or terminates at, the igneous complex of the Selkirk Crest [Miller and Engels, 1975].

The Lewis and Clark line itself may skirt the southern edge of the Spokane dome and continue westward beneath the Miocene basalt, possibly connecting with the northwest trending Pasayten fault or other faults to the west. It is also possible that the zone goes through the dome but has been diffused by ductile flowage of material within the interior of the dome.

The Pasayten fault is a steeply southwest dipping zone of gouge, mylonite, and breccia and has associated with it cataclastic zones up to 1.5 km wide along the east side of the fault [Lawrence, 1978]. Lawrence [1978] relates the formation of the cataclastic zones to left-lateral movement in the early Late Cretaceous and suggests that early Cenozoic movement was primarily dip slip.

The right-lateral Straight Creek fault offsets crystalline rocks of the Cascades 200 km and was demonstrably active during the Eocene, according to Misch [1977]. The fault continues northward into Canada as the Fraser fault, apparently offsetting both the Entiat fault and the northern continuation of the Pasayten fault, then terminates through transfer of right lateral movement to the Tintana-Northern Rocky Mountain trench fault zone through a zone of ductile spreading [Price, 1981, p. 276].

The Rocky Mountain thrust and fold belt has been described by Price [1981, p. 268] as an area in which miogeoclinal Precambrian rocks of the western margin of the craton, and their overlying cover of platformal and foreland basin rocks have been "detached from underlying crystalline basement complex along a myriad array of interfingering and overlaping, listric, west dipping thrust faults." The thrust belt spans the entire 800-km-wide Belt terrane north of the Lewis and Clark line, and right-lateral slippage on the line was accompanied by clockwise rotation on the thrusts [Harrison et al., 1980].

Mudge and Earhart's [1980] study of the Lewis thrust and the Hoadley thrust (Figure 3) provides a quantitative estimate of the amount of this rotation. The Lewis thrust and Hoadley thrust, in map view, resemble the outline of the cutting edges of a wide open east facing pair of scissors, overlapping at a hingepoint (Figure 3) in northwestern Montana. Eastward displacement on the Lewis thrust increases from zero near the hingepoint to 65 km at a point along the trace some 100 km to the northwest [Mudge and Earhart, 1980, p. 13]. These authors postulate 30° of clockwise rotation of the Lewis thrust plate, probably during the very late Paleocene and/or very early Eocene. Displacement on the Hoadley thrust appears to be zero near the hingepoint and increases to 70 km eastward at a point some 100 km to the southeast [Mudge and Earhart, 1980, p. 15]. Movement, then, was scissorlike, closing to the east.

The Hoadley thrust is probably not part of the boundary of the Sanpoil block, but it and other parts of the thrust belt in Montana and Wyoming may form the eastern boundary of other blocks like the Sanpoil. For example, Smedes [1958], anticipating some of the conclusions developed here, had proposed that a block encompassing the Idaho batholith had moved eastward during the Laramide orogeny, producing a left-lateral tear zone (the faults of the Lewis and Clark line) along its northern border, and eastward deflections of the fold and thrust belt. Possibly all of the eastwardly convex deflections (the salients of Buetner [1977]) in the sinuous trace of the thrust belt mark the northeastern corners of similarly rotated basement blocks.

The chief problem with this hypothesis is one of timing. The late early Eocene and/or middle Eocene (50 to 40 m.y. B.P.) rotation of the Sanpoil block may not coincide with the age of thrusting, generally perceived to be Late Cretaceous and Paleocene. It is now accepted that thrusts in the western part of the Rocky Mountain thrust belt are commonly older than thrusts to the east [Armstrong and Oriel, 1965; Bally et al., 1966]. Thrusts in west-central and southwestern Montana in this western part of the thrust belt are Late Cretaceous, based on contact relations with plutonic rocks [Ruppel et al., 1981]. One major thrust fault in the

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east-central part of the belt is intruded by a sill of quartz monzonite said to be 57.3 m.y. old [U.S. Geological Survey, 1971] on the basis of unpublished potassium-argon data. However, thrusts at the eastern edge of the thrust belt in Montana cut Paleocene rocks [Schmidt, 1978; Mudge and Earhart, 1980], but uppermost Eocene or lowermost Oligocene strata, though cut by listric faults, are not affected by the thrust faults [Jones, 1969]. According to Jones [1969], the youngest beds affected by the thrusting in Alberta are of middle and possibly late Paleocene age. Upper plate rocks in that area were thrust eastward 40 km, and perhaps more, between Paleocene and late Eocene to Oligocene time [Bally et al., 1966, p. 371].

Price [1980] suggested that the Eocene rotation was in part accommodated by right-lateral movement on faults within the northern Rocky Mountain trench. Reported right-lateral displacement on these faults is substantial in northern British Columbia, but displacement apparently diminishes to the southeast and in southern British Columbia faults within the trench are chiefly normal, reverse, or thrust [Leech, 1966].

To summarize, the discordant paleomagnetic poles of the volcanic rocks suggest about 25° of clockwise rotation of the Sanpoil block between 50 and 40 m.y. B.P. The rotation followed early phases of extension of the block through formation of the grabens and crustal necking within zones in which the metamorphic core complexes are now exposed. We do not know the relative ages of block rotation and final movement on low-angle mylonite zones and faults bounding the east side of the Kettle gneiss dome and west side of the Okanogan gneiss dome. However, we doubt that much of the rotation can be accounted for by displacement on the mylonite zone at the eastern side of the Kettle gneiss dome because rotation of our localities east and west of this feature are comparable. Near the 49th parallel, the boundary between rotated areas of western North America and the unrotated craton appears to be the Rocky Mountain thrust belt. We cannot rule out the possibility that another boundary or boundaries lie between our study area and the thrust belt, separating areas which rotated clockwise at different times. We think it more likely, however, that final movement on the eastern thrusts of the Rocky Mountain thrust belt at this latitude was during the late early or middle Eocene, somewhat later than the Paleocene age commonly assigned to this event. Hence we suggest that as the Sanpoil block rotated clockwise, perhaps in concert with rotation of other blocks to the south [Hamilton, 1981], its northern and eastern margins were thrust northeastward over the craton in a paroxysmal stage of Laramide thrusting.

Speculations on driving mechanisms. With the exception of Siletzia, tectonostratigraphic terranes west of the study area were approximately in place prior to the Eocene [Monger et al., 1982, Figure 2]. These terranes include areas west of the Ross Lake-Hozameen fault which on the basis of paleomagnetic evidence probably had a middle and Late Cretaeous location far to the south [Beck et al., 1982].

The early Cenozoic disruption of these terranes involved both dextral shear, as evidenced by the 200 km right-lateral offset on the Straight Creek fault [Misch, 1977] and at a later stage, northeastward directed compression. The formation of these features was probably related in some way to the broadly coeval plate rupture, crustal necking, and volcanism within the study area and to the docking, rotation, and possible northward transport of Siletzia [Fox, 1983, p. 32].

Paleomagnetic declination of Paleocene and lower Eocene basement rocks of Siletzia show that they have rotated 25°-75° clockwise relative to the North American craton [Simpson and Cox, 1977; Beck, 1980; Globerman et al., 1982; Magill et al., 1981; Wells and Coe, 1980]. Paleomagnetic inclinations of these rocks are very nearly concordant. However, there is a marked tendency for inclinations to be slightly too low [Bates et al., 1981]; the average flattening of inclination for seven separate exposures of Eocene rocks of Siletzia is 3.7° (data from Beck [1980], Magill et al. [1981], Globerman et al. [1982], and Wells and Coe [1980]). This probably means that Siletzia moved northward perhaps 300-400 km during and/or following accretion. This in turn suggests that northward movement and dextral shear of terranes composing the western margin of the continent in Oregon, Washington, and

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British Columbia did not end until after 49 m.y. ago.

It seems likely that the rotation of the Sanpoil block was related to the rotation and northward translation of Siletzia and, hence, to the interaction of North America with one or more oceanic plates off its western margin [Fox, 1983, p. 36; Engebretson, 1982].

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